

Neil Hasenstaub

St. Petersburg, FL 33705 • nhasenstaub5366@floridapoly.edu • 727-259-9965 • <https://neilh9056.github.io>

Education

Bachelor of Science in Mechanical Engineering, Advanced Topics. GPA: 4.0 Expected May 2027
Florida Polytechnic University Lakeland, FL
Relevant Coursework: Structure and Properties of Materials, Kinematics and Dynamics of Machinery, Mechatronic Systems, Fluid Mechanics, Intro to Aero Structures, Mechanical Lab Design, Heat Transfer

Engineering Projects

Orthopedic Surgical Power Instrument

Shukla Medical Jun 2026 – Aug 2026

- Designed a modular electromechanical interface for a prototype power surgical instrument
- Incorporated DFM principles to model CNC-machinable components in SolidWorks
- Collaborated with engineers to combine subsystems while considering surgical environment conditions

Inertial Dynamometer Laboratory Instrument

Mechanical Lab Design 2 Jan 2026 – May 2026

- Designed an inertial dynamometer system featuring universal motor mounting and object inertia testing
- Implemented a Hall-effect tachometer with Arduino to measure rotational speed over time
- Developed a comprehensive software package and operations manual for gathering and processing data

Four-Link Mechanism Analysis

Kinematics and Dynamics of Machinery Sep 2025 – Dec 2025

- Compared four-link mechanism designs to optimize approach to a logistics problem
- Utilized MATLAB-implemented kinematics and dynamics to examine mechanism behavior under loading
- Performed FEA in SolidWorks to preview expected stress/strain and locate stress concentrations

Autonomous Vehicle Design

CAT Robotics Capstone Aug 2022 – May 2023

- Designed and 3D-printed custom vehicle chassis for modular 4WD independent suspension system
- Integrated SONAR rangefinder and infrared sensors to implement obstacle avoidance and line following
- Applied Arduino C++ to create PPM-controlled mode selection and LCD information display

Leadership & Activities

Music Club E-Board Sep 2026 – Present

- Assisted with event setup, club communication, and interclub relations

Lab Assistant – Chemistry 1 Aug 2024 – Apr 2025

- Setup & teardown of lab experimental materials and assisted in conducting student experiments

ASME Member Sep 2023 – Feb 2024

- Designed airless tires with improved treads and mounting for club lunar rover project

Skills

Technical: Advanced SolidWorks (CSWP) CAD Modeling & Assembly, Fusion 360, Inventor, Excel
Intermediate MATLAB, Java, C/C++, Arduino, Python, C#, RStudio, LabVIEW, AutoCAD

Manufacturing: 3D Printing/Slicing, Technical Drawings, GD&T, FEA (SolidWorks), Simcenter Femap NASTRAN

Interests

- Avid musician who enjoys playing guitar, bass, and drums and collaborating in band settings
- Building and modifying LEGO Technic models, experimenting with different mechanisms for the challenge
- Fan of hockey and baseball, following game strategy and statistics